

Sai Navyanth Penumaka

✉ sp8138@nyu.edu

📄 in/sai-navyanth-p

☎ +1 (704) 726-6905

🌐 sai-navyanth-p.github.io

🔗 [sai-navyanth-p](#)

Education

New York University

New York City, United States

Master of Science in Computer Engineering, **Grade: 4.0/4.0**

Aug 2024 - Dec 2025

– **Teaching Assistant** for ‘Intro to Computer Science’ (Java, Data Structures, Object Oriented Programming)

– **Courses Enrolled:** Machine Learning, Deep Learning, MLOps, Big Data, Computer Architecture

Indian Institute of Technology (IIT) Bhubaneswar

Bhubaneswar, India

Bachelor of Technology (**Honors**) in Electrical Engineering, **Grade: 3.77/4.0**

Jul 2018 - May 2022

Technical Skills

Programming: C++, Java, Python, C, Scala, MATLAB, SQL

Core Competencies: Software Development, Distributed Systems, System Design, Databases, Testing, Deep Learning

Web Development: 1) **Front-end:** HTML, CSS, Javascript, React.js; 2) **Back-end:** REST APIs, Database Systems

Technologies/Libraries: Elasticsearch, Kafka, Docker, Apache Spark, Kubernetes, Hibernate, Linux, Google Guice

Others: Git, MySQL, PostgreSQL, Spring, Django, Node.js, CI/CD, APIs, OOPs, HLD, LLD, Dropwizard, Unix, Azkaban

AI Skills: Machine Learning, Neural Networks, Natural Language Processing, Keras, Numpy, Sklearn, TensorFlow, PyTorch

Work Experience

Flipkart Internet Private Limited (Walmart)

Bengaluru, India

Software Development Engineer (26 months)

Jun 2022 - Aug 2024

- Architected and coded critical back-end services (Offer & Price recommendation), contributing significantly to **\$ 7B**
- Spearheaded the revamp of the offers system, where sellers can opt-in, modify, and opt-out of offers. Designed and programmed APIs, resulting in a **50% increase** in offer opt-ins and a **75% reduction** in **API latencies**
- Successfully upgraded a critical MySQL cluster from version 5.6 to 5.7 and optimized tables, reducing disk space usage from **91% to 30%**, **CPU utilization** from **17% to 5%**, and **latency** by **50%**.
- Spearheaded the creation of ‘Seller Penalization System’ from conception to implementation using **Java, Elasticsearch, Kafka, and Apache Spark**, achieving a substantial **10% reduction** in returns and enhancing product quality
- **Single-handedly** engineered and developed the ‘rewards-serving’ service from the **ground up** using Java, enabling sellers to manage rewards through a dashboard, achieving a **400% return** (RPC = 5)
- Architected rewards-serving through high-level (HLD) and low-level design (LLD), built **Dropwizard Java** Application powered by **Elasticsearch** database, engineered **APIs**, and set up **CI/CD** pipelines and **Kubernetes** infrastructure
- Automated troubleshooting of seller escalations and system issues by creating a self-service dashboard, reducing manual resolution **time by 80%**. Notably, automation of MySQL archival reduced **KTLO by 100%**
- Created price-recommendation models using **Spark & Scala**, boosting weekly adoptions from **93K to 246K (165%)**

Software Engineer Intern (3 months)

May 2021 - Jul 2021

- Reduced Scribe’s **CPU utilization** from **70% to 14%** and increased HTML to PDF rendering **speed by 3 times**
- Implemented and conducted load testing on 3 solutions, each yielding a minimum **40% reduction** in CPU utilization

Key Projects

RoBERTa LoRA Text Classifier | Python, PyTorch, Hugging Face, LoRA | Natural Language Processing

[Github](#)

- Achieved 88.1% validation accuracy on AG News by fine-tuning only 0.5% of RoBERTa parameters (667K/125M) with LoRA, reducing compute and memory by 99%+ through a modular, resource-efficient pipeline that outperformed full fine-tuning under a strict 1M parameter budget.

Course Registration System | Java, Web Development

[Github](#)

- Designed and developed a robust **RESTful** Web Application using **Java** and **MySQL** adhering to **SOLID** principles
- Students can add, delete, or drop courses and check grades while professors choose courses to teach and submit grades

Detection of COVID-19 from Chest X-Rays using Machine Learning | Research | IIT Bhubaneswar

[Thesis](#)

- Implemented a computationally efficient and highly accurate Machine Learning model (Convolutional Neural Network) capable of detecting COVID-19 from chest X-ray images, achieving a remarkable **accuracy of 97.17%**
- Conducted a comprehensive performance evaluation of SVM, KNN, ANN, CNN, VGG 16, DenseNet 121 and Xception

Green Wallet AI | Python, Flask, MongoDB, Machine Learning, Gen AI, Llama LLM

[Hack NYU 2025](#)

- Built a Python/Flask web app in 24 hours to track eco-friendly spending and calculate real-time rewards; integrated an AI assistant (Llama LLM) to analyze 30-day transaction history, answer user queries, and deliver personalized insights

Awards/Achievements/Leadership

- Awarded prestigious ‘**Instant Karma**’ **Employee Achievement Award** for outstanding contributions, Q4 2022
- Achieved **3rd place** in the company-wide **Hackathon** for building a Price Recommendation Model using ML
- **General Secretary** | **Alumni Cell and International Affairs**, IIT Bhubaneswar | 2021-2022